# New Wheels Project

## Introduction to SQL

## Problem Statement

### Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

### Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

### Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

**Solution Query:**

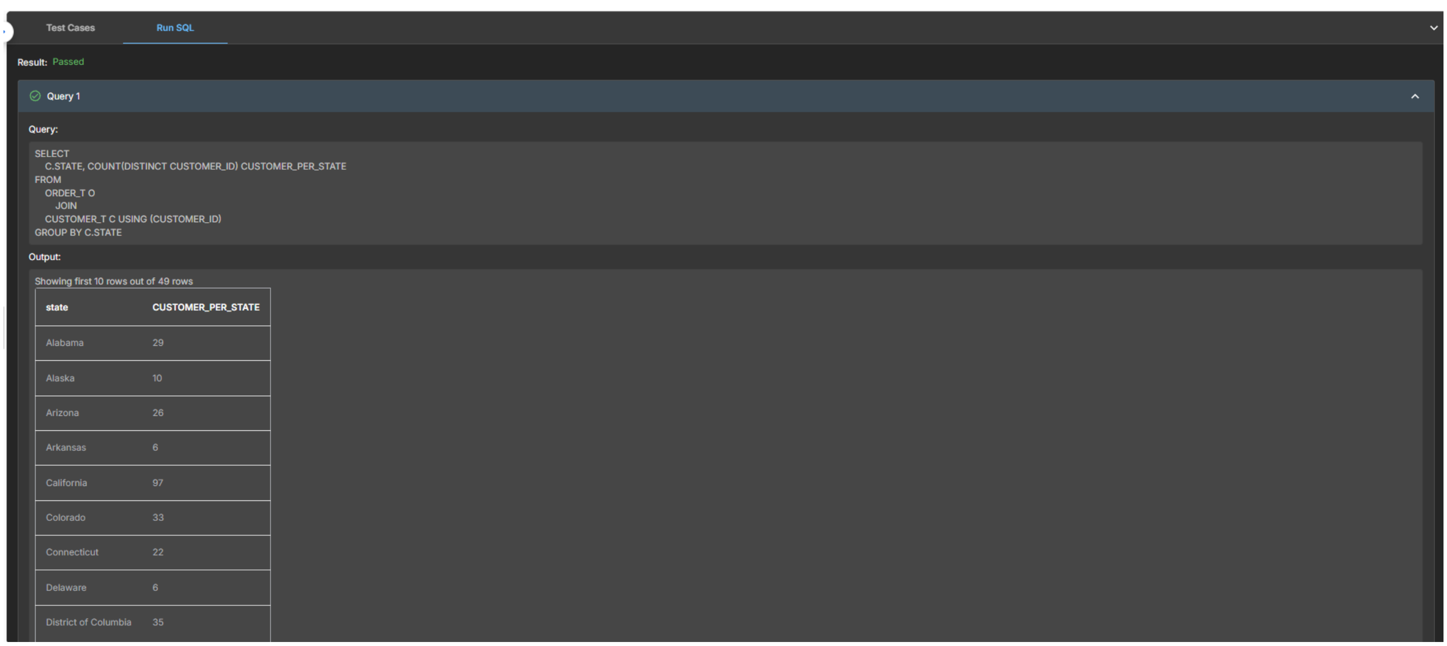
```
-- Total number of customer

SELECT
    COUNT(DISTINCT CUSTOMER_ID) TOTAL_NO_OF_CUSTOMER
FROM
    ORDER_T O
        JOIN
    CUSTOMER_T C USING (CUSTOMER_ID);


-- distribution of the customers across states

SELECT
    C.STATE, COUNT(DISTINCT CUSTOMER_ID) CUSTOMER_PER_STATE
FROM
    ORDER_T O
        JOIN
    CUSTOMER_T C USING (CUSTOMER_ID)
GROUP BY C.STATE;
```

**Output:**



**Observations and Insights:**

- The total number of customers who have placed orders are 994.
- The highest number of orders i.e. 97 orders were placed from the California and Texas state.
- There are total 49 state from where the customers belong who placed the order.

## Question 2: Which are the top 5 vehicle makers preferred by the customers?

**Solution Query:**

```
SELECT
   P.VEHICLE_MAKER, COUNT(CUSTOMER_ID) NO_OF_CUSTOMER
FROM
   ORDER_T O
      JOIN
   CUSTOMER_T C USING (CUSTOMER_ID)
      JOIN
   PRODUCT_T P USING (PRODUCT_ID)
GROUP BY P.VEHICLE_MAKER
ORDER BY 2 DESC
LIMIT 5;
```

**Output:**

Test Cases    Run SQL

Result: Passed

Query 1

Query:

```
SELECT
   P.VEHICLE_MAKER, COUNT(CUSTOMER_ID) NO_OF_CUSTOMER
FROM
   ORDER_T O
      JOIN
   CUSTOMER_T C USING (CUSTOMER_ID)
      JOIN
   PRODUCT_T P USING (PRODUCT_ID)
GROUP BY P.VEHICLE_MAKER
ORDER BY 2 DESC
LIMIT 5
```

Output:

Showing 5 rows

| vehicle_maker | NO_OF_CUSTOMER |
|---|---|
| Chevrolet | 83 |
| Ford | 63 |
| Toyota | 52 |
| Pontiac | 50 |
| Dodge | 50 |

**Observations and Insights:**

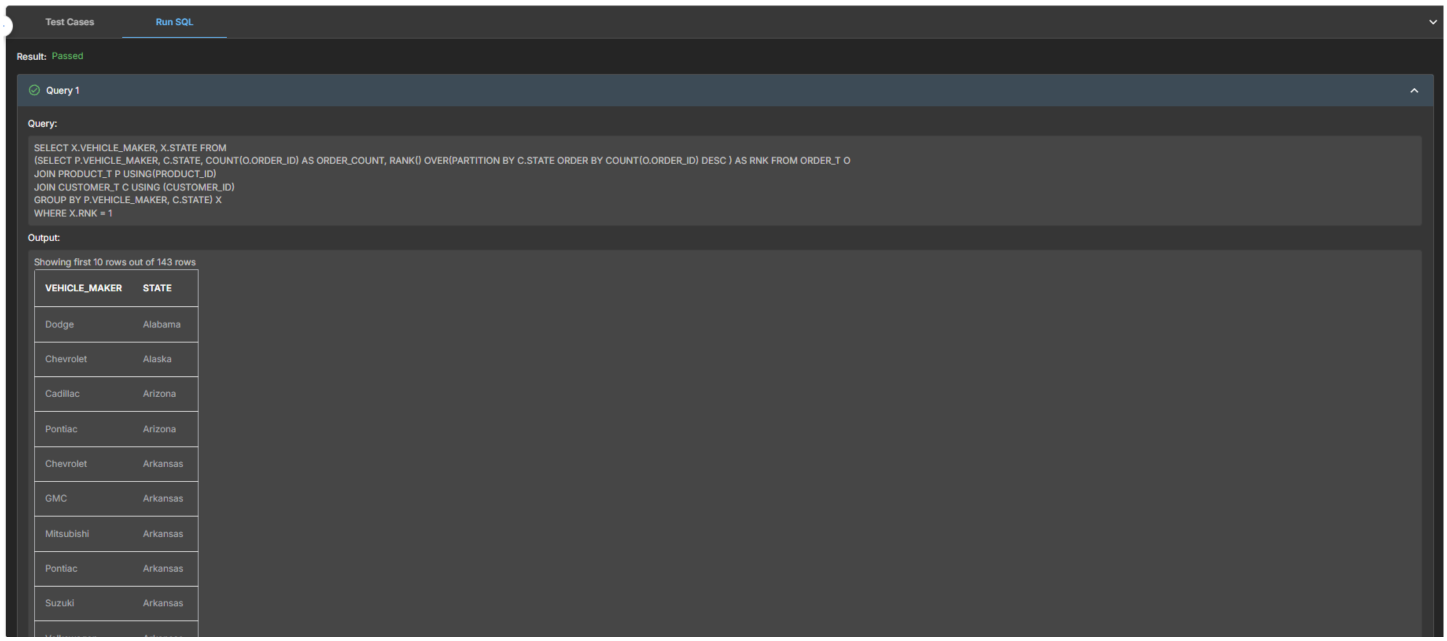
- **The top 5 vehicle makers preferred by the customers are Chevrolet, Ford, Toyota, Pontiac and Dodge.**
- **There are total** 54 **vehicle makers.**
- There are lots of **vehicle makers that are least preferred by the customers are Austin, Ram, MG, etc.**

## Question 3: Which is the most preferred vehicle maker in each state?

**Solution Query:**

```
SELECT X.VEHICLE_MAKER, X.STATE FROM
(SELECT P.VEHICLE_MAKER, C.STATE, COUNT(O.ORDER_ID) AS ORDER_COUNT, RANK()
OVER(PARTITION BY C.STATE ORDER BY COUNT(O.ORDER_ID) DESC ) AS RNK FROM ORDER_T O
JOIN PRODUCT_T P USING(PRODUCT_ID)
JOIN CUSTOMER_T C USING (CUSTOMER_ID)
GROUP BY P.VEHICLE_MAKER, C.STATE) X
WHERE X.RNK = 1;
```

**Output:**


Test Cases
Run SQL
Result: Passed
Query 1
Query:

```
SELECT X.VEHICLE_MAKER, X.STATE FROM
(SELECT P.VEHICLE_MAKER, C.STATE, COUNT(O.ORDER_ID) AS ORDER_COUNT, RANK() OVER(PARTITION BY C.STATE ORDER BY COUNT(O.ORDER_ID) DESC ) AS RNK FROM ORDER_T O
JOIN PRODUCT_T P USING(PRODUCT_ID)
JOIN CUSTOMER_T C USING (CUSTOMER_ID)
GROUP BY P.VEHICLE_MAKER, C.STATE) X
WHERE X.RNK = 1
```

Output:
Showing first 10 rows out of 143 rows

| VEHICLE_MAKER | STATE |
|---|---|
| Dodge | Alabama |
| Chevrolet | Alaska |
| Cadillac | Arizona |
| Pontiac | Arizona |
| Chevrolet | Arkansas |
| GMC | Arkansas |
| Mitsubishi | Arkansas |
| Pontiac | Arkansas |
| Suzuki | Arkansas |

**Observations and Insights:**

- **Chevrolet, Toyota, and Ford are the “big three”** in terms of widespread state-level preference. They consistently appear across multiple regions, suggesting strong national brand loyalty.
- **Ford has broad appeal**: Shows up in California, Hawaii, Illinois, Iowa, Kansas, Louisiana, Maryland, Michigan, North Dakota, Virginia, Wisconsin. Ford’s presence is widespread, especially in the Midwest and East.
- **State diversity**: Some states (like Arkansas, Iowa, Kansas, Utah, Wisconsin) list multiple makers, indicating no single dominant preference — customer choices are fragmented.

**Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?**

**Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5**

**Solution Query:**

```
SELECT
    O.QUARTER_NUMBER,
    ROUND((SUM(CASE
                WHEN O.CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*)),
            2) AS VERY_BAD_PERCENT,
    ROUND((SUM(CASE
                WHEN O.CUSTOMER_FEEDBACK = 'Bad' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*)),
            2) AS BAD_PERCENT,
    ROUND((SUM(CASE
                WHEN O.CUSTOMER_FEEDBACK = 'Okay' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*)),
            2) AS OKAY_PERCENT,
    ROUND((SUM(CASE
                WHEN O.CUSTOMER_FEEDBACK = 'Good' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*)),
            2) AS GOOD_PERCENT,
    ROUND((SUM(CASE
                WHEN O.CUSTOMER_FEEDBACK = 'Very Good' THEN 1
                ELSE 0
            END) * 100.0 / COUNT(*)),
            2) AS VERY_GOOD_PERCENT
FROM
    ORDER_T O
GROUP BY O.QUARTER_NUMBER
ORDER BY O.QUARTER_NUMBER;
```

**Output:**

Test Cases    Run SQL

Result: Passed

Query 1

Query:

```
SELECT
  O.QUARTER_NUMBER,
  ROUND(AVG(CASE
    WHEN O.CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
    WHEN O.CUSTOMER_FEEDBACK = 'Bad' THEN 2
    WHEN O.CUSTOMER_FEEDBACK = 'Okay' THEN 3
    WHEN O.CUSTOMER_FEEDBACK = 'Good' THEN 4
    WHEN O.CUSTOMER_FEEDBACK = 'Very Good' THEN 5
  END),2) AS OVERALL_AVG_RATING
FROM
  ORDER_T O
GROUP BY O.QUARTER_NUMBER
ORDER BY O.QUARTER_NUMBER
```

Output:

Showing 4 rows

| quarter_number | OVERALL_AVG_RATING |
|---|---|
| 1 | 3.55 |
| 2 | 3.35 |
| 3 | 2.96 |
| 4 | 2.4 |

**Observations and Insights:**

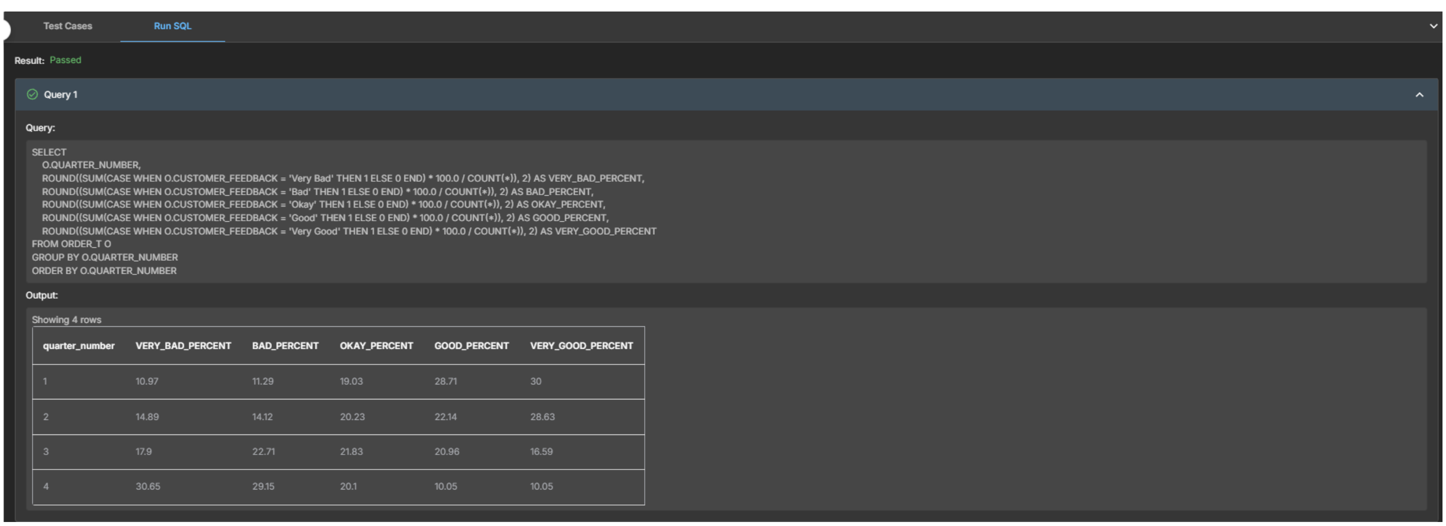
- **Q1: 3.55** → Customers were relatively satisfied, with ratings above the midpoint (closer to "Good" than "Okay").
- **Q2: 3.35** → Slight decline compared to Q1, but still in the "Okay–Good" range.
- **Q3: 2.96** → Noticeable drop, dipping below 3.0, which suggests customers leaned more toward "Okay" or "Bad."
- **Q4: 2.40** → Significant decline, showing dissatisfaction increased sharply by year-end.

## Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

**Solution Query:**

```
SELECT
   O.QUARTER_NUMBER,
   (SUM(CASE
      WHEN O.CUSTOMER_FEEDBACK = 'Very Bad' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS 'VERY_BAD(%)',
   (SUM(CASE
      WHEN O.CUSTOMER_FEEDBACK = 'Bad' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS 'BAD(%)',
   (SUM(CASE
      WHEN O.CUSTOMER_FEEDBACK = 'Okay' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS 'Okay(%)',
   (SUM(CASE
      WHEN O.CUSTOMER_FEEDBACK = 'Good' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS 'Good(%)',
   (SUM(CASE
      WHEN O.CUSTOMER_FEEDBACK = 'Very Good' THEN 1
      ELSE 0
   END) / COUNT(*)) * 100 AS 'Very Good(%)'
FROM
   ORDER_T O
GROUP BY O.QUARTER_NUMBER
ORDER BY O.QUARTER_NUMBER;
```

**Output:**


Test Cases    Run SQL

Result: Passed

Query 1

Query:

```
SELECT
   O.QUARTER_NUMBER,
   ROUND((SUM(CASE WHEN O.CUSTOMER_FEEDBACK = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS VERY_BAD_PERCENT,
   ROUND((SUM(CASE WHEN O.CUSTOMER_FEEDBACK = 'Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS BAD_PERCENT,
   ROUND((SUM(CASE WHEN O.CUSTOMER_FEEDBACK = 'Okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS OKAY_PERCENT,
   ROUND((SUM(CASE WHEN O.CUSTOMER_FEEDBACK = 'Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS GOOD_PERCENT,
   ROUND((SUM(CASE WHEN O.CUSTOMER_FEEDBACK = 'Very Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS VERY_GOOD_PERCENT
FROM ORDER_T O
GROUP BY O.QUARTER_NUMBER
ORDER BY O.QUARTER_NUMBER
```

Output:

Showing 4 rows

| quarter_number | VERY_BAD_PERCENT | BAD_PERCENT | OKAY_PERCENT | GOOD_PERCENT | VERY_GOOD_PERCENT |
|---|---|---|---|---|---|
| 1 | 10.97 | 11.29 | 19.03 | 28.71 | 30 |
| 2 | 14.89 | 14.12 | 20.23 | 22.14 | 28.63 |
| 3 | 17.9 | 22.71 | 21.83 | 20.96 | 16.59 |
| 4 | 30.65 | 29.15 | 20.1 | 10.05 | 10.05 |

**Observations and Insights:**

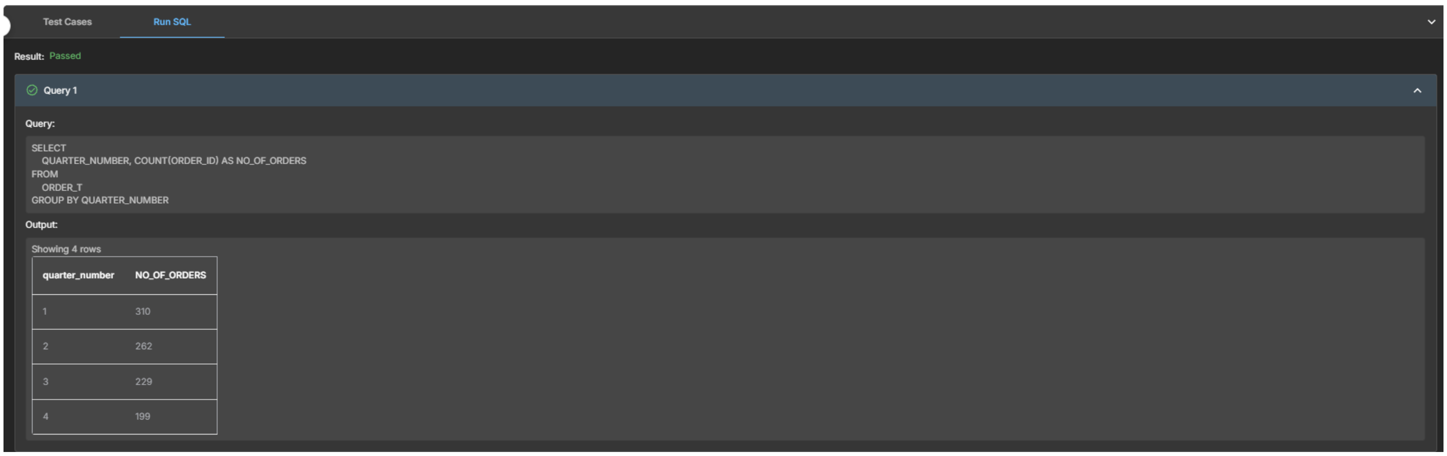
- **Clear deterioration**: Customer sentiment worsened quarter by quarter, mirroring the decline in orders, revenue, and shipping performance.
- **Positive feedback collapse**: “Very Good” fell from 30% in Q1 to just 10% in Q4 — a two-thirds reduction.
- **Correlation with operations**: The rise in shipping delays (58 → 175 days) aligns directly with worsening feedback.
- **Business risk**: By Q4, the majority of customers were unhappy, which explains the sharp drop in orders and revenue.

## Question 6: What is the trend of the number of orders by quarter?

**Solution Query:**

```
SELECT
   QUARTER_NUMBER, COUNT(ORDER_ID) AS NO_OF_ORDERS
FROM
   ORDER_T
GROUP BY QUARTER_NUMBER;
```

**Output:**




**Observations and Insights:**

- **Consistent Decline**: Orders decreased quarter after quarter, showing a clear negative trend across the year.
- **Demand Weakening**: The drop from Q1 (310) to Q4 (199) is about **36% overall**, which is significant.
- The company needs to investigate why early momentum wasn't sustained.

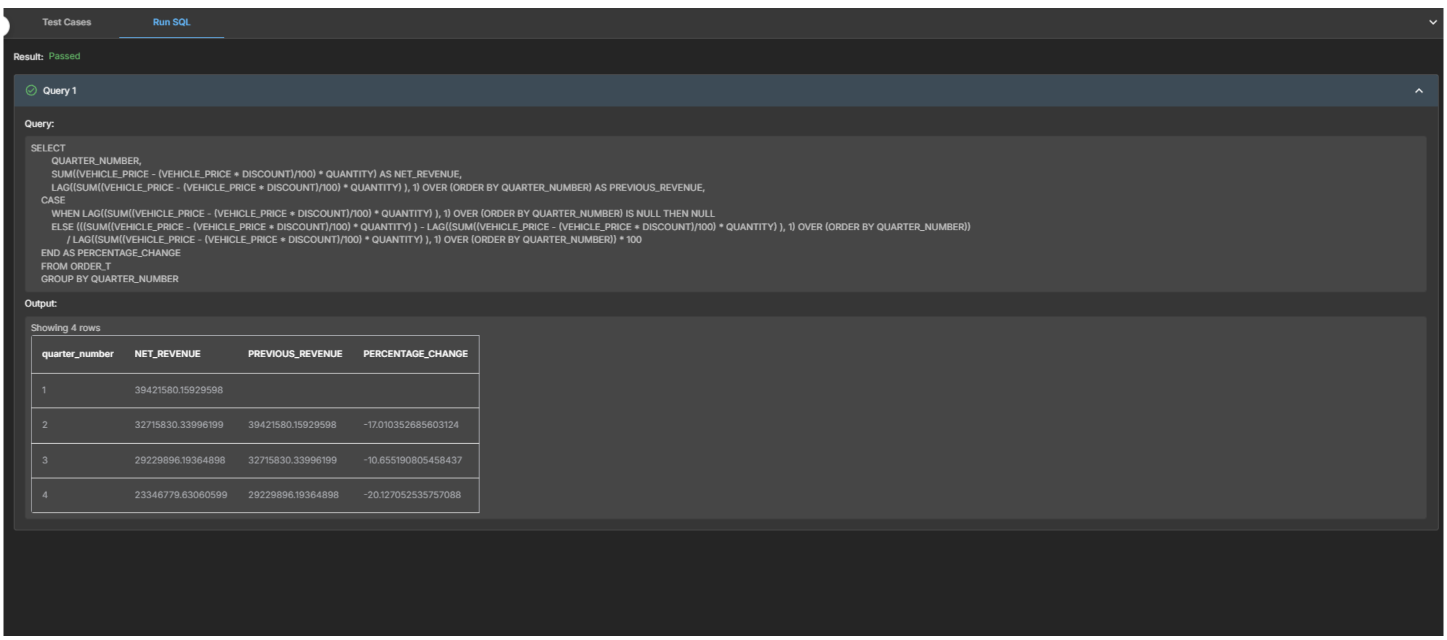
 

## Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

**Solution Query:**

```
WITH revenue_per_quarter AS (
    SELECT
        QUARTER_NUMBER,
        SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) AS NET_REVENUE
    FROM ORDER_T
    GROUP BY QUARTER_NUMBER
)
SELECT
    QUARTER_NUMBER,
    NET_REVENUE,
    LAG(NET_REVENUE, 1) OVER (ORDER BY QUARTER_NUMBER) AS PREVIOUS_REVENUE,
    CASE
        WHEN LAG(NET_REVENUE, 1) OVER (ORDER BY QUARTER_NUMBER) IS NULL THEN NULL
        ELSE ((NET_REVENUE - LAG(NET_REVENUE, 1) OVER (ORDER BY QUARTER_NUMBER))
              / LAG(NET_REVENUE, 1) OVER (ORDER BY QUARTER_NUMBER)) * 100
    END AS PERCENTAGE_CHANGE
FROM revenue_per_quarter
ORDER BY QUARTER_NUMBER;
```

**Output:**


Test Cases Run SQL

Result: Passed

Query 1

Query:

```
SELECT
    QUARTER_NUMBER,
    SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) AS NET_REVENUE,
    LAG((SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) ), 1) OVER (ORDER BY QUARTER_NUMBER) AS PREVIOUS_REVENUE,
  CASE
    WHEN LAG((SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) ), 1) OVER (ORDER BY QUARTER_NUMBER) IS NULL THEN NULL
    ELSE (((SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) ) - LAG((SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) ), 1) OVER (ORDER BY QUARTER_NUMBER))
        / LAG((SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT)/100) * QUANTITY) ), 1) OVER (ORDER BY QUARTER_NUMBER)) * 100
  END AS PERCENTAGE_CHANGE
  FROM ORDER_T
  GROUP BY QUARTER_NUMBER
```

Output:

Showing 4 rows

| quarter_number | NET_REVENUE | PREVIOUS_REVENUE | PERCENTAGE_CHANGE |
|---|---|---|---|
| 1 | 39421580.15929598 | | |
| 2 | 32715830.33996199 | 39421580.15929598 | -17.010352685603124 |
| 3 | 29229896.19364898 | 32715830.33996199 | -10.655190805458437 |
| 4 | 23346779.63060599 | 29229896.19364898 | -20.127052535757088 |

**Observations and Insights:**

- **Consistent Decline: Net revenue fell every quarter, with no recovery phase.**
- **Sharpest Drop in Q4**: The -20% decline in Q4 is the steepest, signaling worsening business conditions.

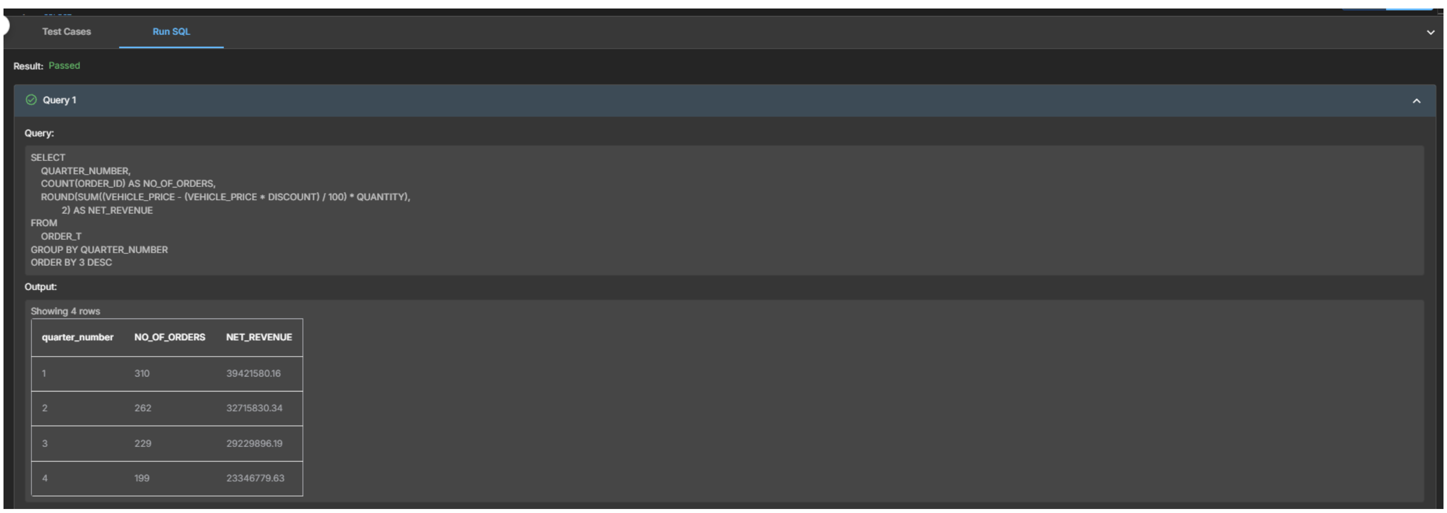
- **Overall Impact**: Revenue dropped from ₹39.42M in Q1 to ₹23.35M in Q4 — a **41% decline across the year**.
- **Urgent intervention needed**: Without corrective action, the downward trajectory will continue.

## Question 8: What is the trend of net revenue and orders by quarters?

**Solution Query:**

```
SELECT
   QUARTER_NUMBER,
   COUNT(ORDER_ID) AS NO_OF_ORDERS,
   ROUND(SUM((VEHICLE_PRICE - (VEHICLE_PRICE * DISCOUNT) / 100) * QUANTITY),
         2) AS NET_REVENUE
FROM
   ORDER_T
GROUP BY QUARTER_NUMBER
ORDER BY 3 DESC;
```

**Output:**




| quarter_number | NO_OF_ORDERS | NET_REVENUE |
|---|---|---|
| 1 | 310 | 39421580.16 |
| 2 | 262 | 32715830.34 |
| 3 | 229 | 29229896.19 |
| 4 | 199 | 23346779.63 |

**Observations and Insights:**

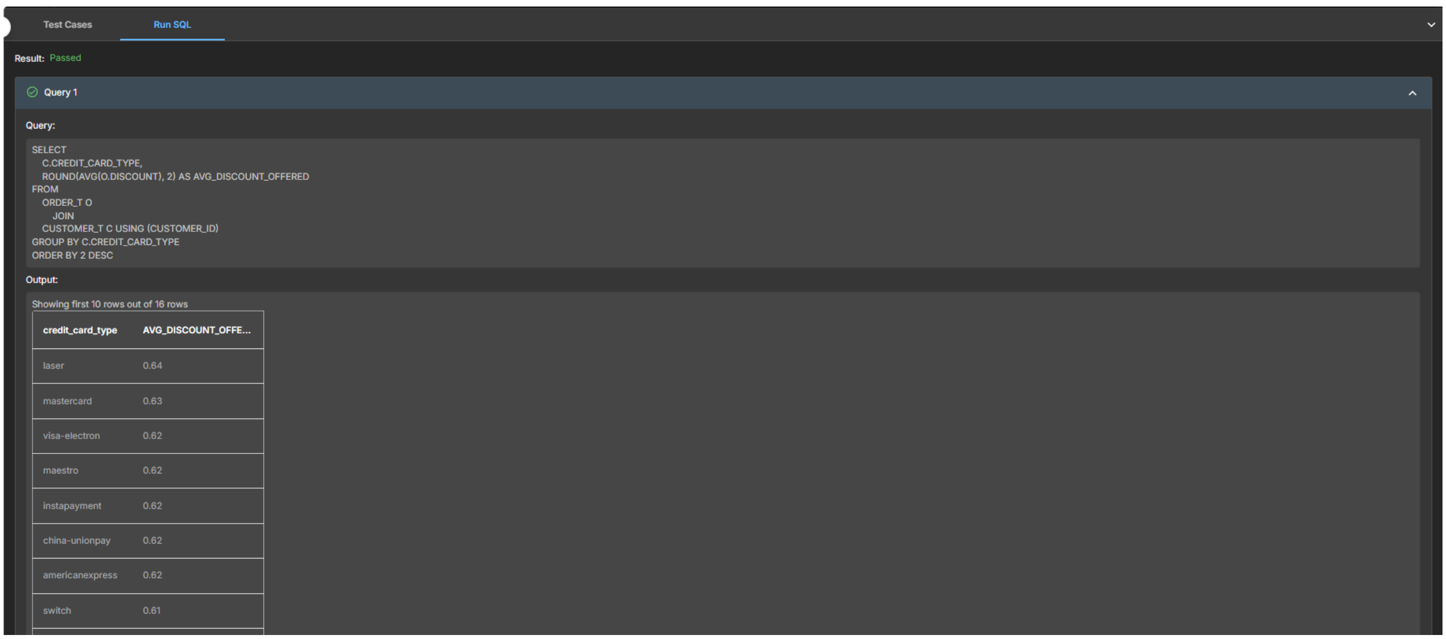
- **Consistent Downward Trend**: Both **orders and revenue decreased quarter-over-quarter**, showing weakening demand and performance.
- **Correlation with Customer Ratings**: Customer satisfaction also dropped (from 3.55 in Q1 to 2.40 in Q4). The parallel decline in ratings, orders, and revenue indicates a strong link between customer sentiment and sales.
- **Critical Q4 Performance**: The sharpest revenue drop (-20%) in Q4 highlights operational or market issues that worsened over time.

## Question 9: What is the average discount offered for different types of credit cards?

**Solution Query:**

```
SELECT
   C.CREDIT_CARD_TYPE,
   ROUND(AVG(O.DISCOUNT), 2) AS AVG_DISCOUNT_OFFERED
FROM
   ORDER_T O
      JOIN
   CUSTOMER_T C USING (CUSTOMER_ID)
GROUP BY C.CREDIT_CARD_TYPE
ORDER BY 2 DESC;
```

**Output:**




**Observations and Insights:**

- **Laser (0.64)** has the highest average discount, slightly ahead of others.
- **Minimal Variation**: The spread between the highest (0.64) and lowest (0.58) average discount is only **0.06**, suggesting the company maintains a fairly uniform discount policy across card types.
- **Laser & Mastercard Advantage**: Customers using **Laser** and **Mastercard** enjoy slightly better discounts, which could be leveraged in marketing campaigns to highlight benefits.

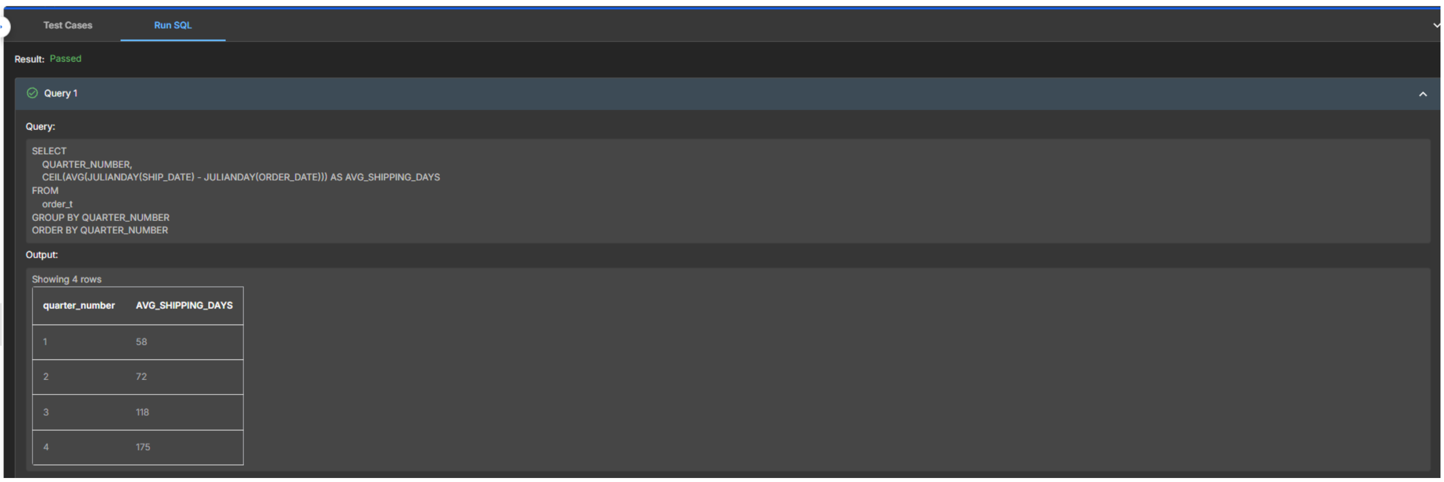
 

## Question 10: What is the average time taken to ship the placed orders for each quarter?

**Solution Query:**

```
SELECT
    QUARTER_NUMBER,
    CEIL(AVG(JULIANDAY(SHIP_DATE) - JULIANDAY(ORDER_DATE))) AS AVG_SHIPPING_DAYS
FROM
    order_t
GROUP BY QUARTER_NUMBER
ORDER BY QUARTER_NUMBER;
```

**Output:**




**Observations and Insights:**

- **Steady Deterioration: Shipping times increased quarter by quarter, with the worst delays in Q4.**
- **Strong Correlation with Customer Ratings: As shipping times grew longer (58 → 175 days), customer ratings fell (3.55 → 2.40). This strongly suggests that delayed deliveries were a major driver of dissatisfaction.**
- **Average shipping times worsened dramatically each quarter, rising from 58 days in Q1 to 175 days in Q4. This trend aligns with falling ratings, orders, and revenue, making logistics performance the critical issue to address.**

# Business Metrics Overview

| Total Revenue | Total Orders | Total Customers | Average Rating |
|---|---|---|---|
| 124714086.32 | 1000 | 994 | 3.14 |
| **Last Quarter Revenue** | **Last quarter Orders** | **Average Days to Ship** | **% Good Feedback** |
| 23346779.63 | 199 | 98 | 21.5 |

# Business Recommendations

- Invest in supply chain optimization, diversify shippers, and introduce stricter SLA monitoring as average shipping time ballooned from 58 days (Q1) to 175 days (Q4). This is the single biggest driver of dissatisfaction. →
- Launch customer feedback programs, resolve complaints faster, and offer compensation for delayed orders as ratings dropped from 3.55 (Q1) to 2.40 (Q4).
- Introduce loyalty programs, seasonal promotions, and personalized offers to re-engage customers.
- Re-evaluate discount strategy to avoid eroding margins.
- The company's **decline in orders, revenue, ratings, and shipping performance are tightly linked**.
- Since vehicle maker preferences vary by state, tailor marketing campaigns regionally (e.g., Toyota in the South, Chevrolet in the Midwest).